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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: BTECH | Year: III Semester: VI

Stream : Cyber Security-CSE

Subject: Digital Forensic and Incident Response | Time: 3 hrs (2pm to 5pm)

Date: 22/04/2024 | No. of Pages : 2

Marks: 100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1 ; SO-2 ; BL-2	a.	Explain what is a chain of custody ?	5
CO-1 ; SO-2 ; BL-2	b.	How can file carving techniques be used in digital forensics investigations?	5
CO-1 ; SO-2 ; BL-2	c.	How does RTO differ from RPO in business impact analysis?	5
CO-1 ; SO-2 ; BL-2	d.	What is the significance of cluster numbers in Microsoft file systems for file fragmentation?	5
Q2 CO-1 ; SO-2 ; BL-3	a.	Explain the difference between direct and indirect evidence in the context of mobile forensics investigations. Provide examples of each and discuss their significance in determining culpability or innocence in legal proceedings.	10
CO-3 ; SO-2 ; BL-2	b.	i) State and explain any two **Indian Penal Code (IPC)** applicable for Cyber crimes ? [5] ii) Explain various Cyber Law applicable in Indian context? [5]	10
Q3 CO-2; SO-2; BL-3	a.	How can network traffic analysis contribute to router forensics investigations?	10
CO-2; SO-2; BL-2	b.	Discuss the key characteristics, challenges, and implications of the dark web for digital forensics investigations.	10

Q4. CO-1; SO-2; BL-2	a.	Explain the role of the file allocation table (FAT) in the structure of a disk drive. How does the FAT facilitate data storage and retrieval, and what are its key components?	10
CO-2; SO-2; BL-2	b.	Explain the role of the incident response methodology in cybersecurity. What are its major components and how do they contribute to effectively managing security incidents?	10
Q5 CO-2; SO-2; BL-2	a.	When should static and dynamic acquisition methods be employed in digital forensics investigations? Explain in detail the techniques used with examples	10
CO-2; SO-2; BL-2	b.	What are the various techniques of data hiding; explain in detail	10
Q6 CO-3; SO-3; BL-2	a.	What are the various kinds of reports that a forensic examiner will need to write? Describe the function and what sort of information each one will contain.	10
CO-1; SO-2; BL-2	b.	What is DNS spoofing, and how does it work? Explain the steps involved in executing a DNS spoofing attack, including the potential consequences for network security. Describe the concept of MAC address spoofing in network communications. Explain how MAC address spoofing attacks are carried out, including the methods used to change MAC addresses and the implications for network security.	10
Q7 CO-1; SO-3; BL-2	a.	Describe the various metrics used in Business Impact Analysis	10
CO-3; SO-4; BL-2	b.	What are the key responsibilities and ethical obligations of expert witnesses in legal proceedings? Discuss the challenges expert witnesses may face in maintaining impartiality and credibility while fulfilling their duty to assist the court in understanding complex technical or scientific matters. Provide examples of ethical dilemmas that expert witnesses may encounter and how they can navigate them effectively.	10